Learning Scene-Grounded Interaction Priors for Scene-Aware Human Motion Prediction

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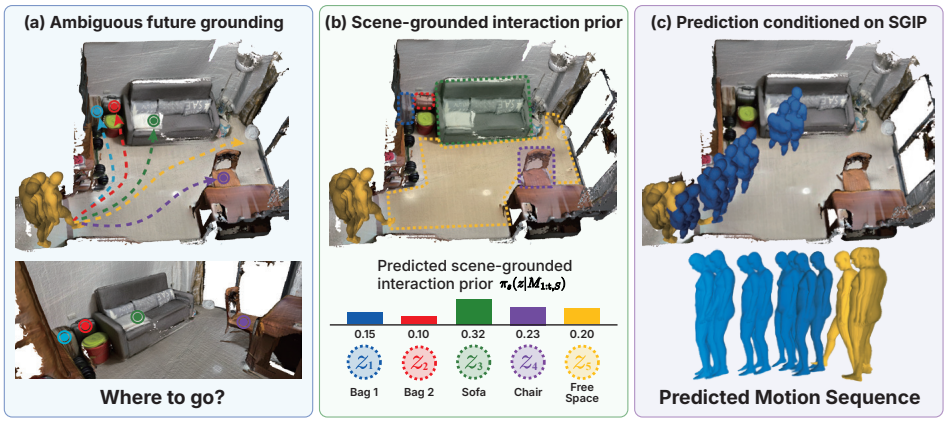


Figure 1: **Overview of the proposed method.** Scene-aware human motion prediction (HMP) requires resolving *future-motion ambiguity*. **(a)** Given the same observed motion, a person may head toward different objects or simply pass through the space: approaching a bag, sitting on the sofa or chair, or freely navigating the room. **(b)** *SGIP* predicts a scene-grounded interaction prior, a distribution over candidate scene tokens (covering both object instances and navigable regions), from the observed motion and 3D scene alone without any auxiliary cues. **(c)** The predicted prior selects a scene token that conditions both trajectory and full-body pose prediction; here, the model correctly anticipates walking toward and sitting on the sofa. Unlike previous methods that resolve such ambiguity *implicitly* or depend on auxiliary cues at deployment, *SGIP* makes scene-interaction reasoning an *explicit, supervisable* target using only the observed motion and 3D scene at inference.

Abstract

Human motion prediction (HMP) aims to forecast future movement from an observed motion sequence, and recent methods have incorporated 3D scene geometry and object semantics to improve long-term plausibility. However, the destination of future motion remains ambiguous in cluttered scenes: the same observed motion can lead to different interaction targets or navigation modes. Existing methods either handle this ambiguity implicitly or rely on auxiliary signals like gaze that are hard to obtain in practice. To address this, we propose *SGIP*, which explicitly predicts where in the scene future motion will be grounded across both objects and open navigation regions, and uses it to condition future trajectory and full-body pose prediction. This scene-grounded interaction prior is learned via intent distillation: *at training time only*, a teacher model scores candidate scene regions by combining an LLM prompted with a textual description of the person's intent with geometric motion-scene features. A student model learns to repro-

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duce these scores from only the observed motion and 3D scene, making *intent and LLM unnecessary at test time*. Across multiple benchmark datasets, SGIP outperforms prior scene-aware baselines on both trajectory and pose accuracy in a realistic deployment setting where neither gaze nor intent is available at inference, demonstrating effectiveness of the proposed framework.

1 Introduction

Human motion prediction (HMP) aims to forecast future human poses and trajectories from an observed motion sequence, and is essential for autonomous systems, embodied agents, and human–robot collaboration [16, 38, 39]. While early approaches largely modeled motion dynamics in isolation [39, 43], recent HMP has shifted toward 3D scenes, where predictions must respect both the physical layout and the semantic affordances of the environment [19, 53]. These scene-aware methods improve long-term plausibility by incorporating human–object context, future contact, and gaze-based intention cues [9, 38, 42, 63].

However, future motion in such scenes is inherently ambiguous: the same observed motion can correspond to multiple valid ways of interacting with or navigating through the scene. For example, a person approaching a dining area may sit on one of several chairs or pass through the space toward another room; each possibility induces a different trajectory, body orientation, and contact pattern. Prior work has attempted to resolve this ambiguity by relying on auxiliary cues such as gaze [38, 63] or manually specified interaction targets [18, 56]. However, such cues require specialized sensors or per-frame supervision and are difficult to obtain in deployment. A practical HMP model should therefore make scene-aware predictions from the observed motion and 3D scene alone.

To address this, we propose SGIP, which makes scene-interaction reasoning an explicit, supervisable target by predicting a scene-grounded interaction prior, a distribution over candidate scene tokens covering both interactive objects and navigable regions. SGIP predicts this prior from the observed motion and 3D scene alone, and uses it to condition future trajectory and full-body pose prediction. However, learning the prior directly from observation is challenging since the observed motion alone often does not specify a clear interaction target. We address this through *privileged intent distillation*. At training time, we leverage a textual description of the person's intent (e.g., "take some rest"), which is practical to obtain through a vision-language model rather than requiring per-sample human annotation. A teacher prompts a large language model with this intent to score how well each candidate scene region aligns with the person's goal. Yet textual intent alone is insufficient when a scene contains multiple instances of the same object class, where the LLM cannot distinguish which specific instance the person is heading toward. We therefore combine the LLM's semantic score with a geometric compatibility score derived from motion–scene distance and orientation, which uses the observed motion to disambiguate between same-class instances. Together, these form the teacher's intent-conditioned distribution over scene tokens. At inference, where neither textual intent nor an LLM is available, a lightweight MLP-based student reproduces this distribution from motion and scene alone, inheriting the teacher's intent-aware scene-interaction reasoning through distillation.

Our contributions are summarized as follows: (1) We introduce a *scene-grounded interaction prior*, an explicit distribution over candidate scene tokens covering both object-centric interactions and free-space navigation, that turns scene-interaction reasoning into a supervisable target. (2) We propose a *privileged intent distillation* framework in which textual intent and an LLM supervise a teacher at training, while the student reproduces the resulting prior from motion and scene alone, requiring no language model or auxiliary cue at inference. (3) On GIMO and PROX, the observation-only student consistently improves trajectory and pose accuracy over scene-aware baselines without auxiliary cues, with the privileged teacher additionally serving as a performance ceiling.

2 Related Work

2.1 Human Motion Prediction

Human motion prediction (HMP) forecasts future human poses or trajectories from observed motion and underpins applications such as autonomous driving, human–robot collaboration, and animation [3, 12, 28, 39, 43, 48]. Architectures span recurrent [12, 43], graph-based [10, 35, 39, 50],

and attention- or Transformer-based [2, 40, 59, 60] formulations, with probabilistic variants (VAEs, flows, GANs, diffusion) modeling multi-future ambiguity [4, 5, 8, 41, 49, 64]. Yet motion-only predictors overlook the surrounding 3D environment and frequently violate scene constraints in deployment. Scene-aware HMP addresses this by injecting scene context, evolving from 2D image- or bounding-box-based formulations [6, 9] to point-cloud fusion [25, 42, 69], typically built on PointNet-style backbones [45, 46] with diverse scene-grounded predictors [13, 54, 58]; gaze further serves as a behavioral proxy for intent in cluttered scenes [38, 63, 69]. Recent methods further reason about potential interaction regions using motion and scene geometry [36], or model human–human and human–scene interactions in dynamic environments [51]. A complementary line synthesizes scene-aware motion conditioned on object affordances [19, 31, 32, 34, 53] or textual action descriptions and activity labels [7, 55, 56], rather than forecasting motion from observed histories.

2.2 Language Models for Scene and Motion Understanding

Large language models (LLMs) have recently been used to inject commonsense knowledge into 3D scene and motion tasks. Motion–language alignment models adapt LLMs to translate textual descriptions into human motion sequences [30], and scene-aware variants further ground the generated motion in 3D environments through textual scene or activity descriptions [7, 55, 56, 62]. Closely related, several methods route language conditions through explicit *affordance* or *intent* representations as intermediate structure between language and motion [11, 15, 57], reflecting a broader shift toward semantically structured intermediates rather than direct text-to-motion mapping. Beyond generation, LLMs serve as reasoners for 3D scene and human–scene understanding by aligning object semantics with language [23, 24, 66], and as grounding modules that connect language goals to spatial affordances for embodied agents [1]. A complementary line of work treats LLMs as judges or rubric-based scorers [68], where natural-language criteria elicit interpretable, score-level outputs over candidate options. Despite this flexibility, deploying LLMs at inference incurs substantial computational and latency cost, which is poorly suited to real-time motion forecasting in deployment.

2.3 Privileged Information and Knowledge Distillation

A natural alternative is to use such expensive or hard-to-obtain supervision only at training time. Learning with privileged information [37, 52] formalizes this setting by exploiting signals available during training but absent at inference, while knowledge distillation [21, 47] transfers such information from a teacher to a more deployable student through output- or intermediate-feature-level imitation. A concrete realization in vision is *modality hallucination* or *modality distillation*, where a privileged modality (e.g., depth or optical flow) supervises a student that operates from a single observable modality at inference [14, 22, 29, 67]. In the context of human motion in 3D scenes, behavioral or semantic cues such as gaze [38, 63, 69], language goals [7, 56], or explicitly stated intent [15] can disambiguate human–scene behavior but are rarely accessible in practical deployment, making them natural candidates for privileged supervision. We adopt this paradigm with two privileged training-time signals—textual intent and an LLM-based rubric scorer—and distill the resulting *scene-grounded interaction prior* into an observation-only student, rather than transferring gaze- or language-conditioned motion outputs as in prior distillation.

3 Proposed Method

SGIP is a two-branch teacher–student framework (Figure 2). The teacher fuses an LLM-based rubric score with a geometric compatibility score into an intent-conditioned interaction prior, and a student approximates this prior from motion and scene alone, serving as the deployed model. Section 3.1 formalizes the problem; Section 3.2 defines the scene-grounded interaction prior and its teacher and student forms; Section 3.3 describes how the prior conditions trajectory and full-body pose prediction; and Section 3.4 describes the distillation procedure. Implementation details (LLM call, prompt templates, hyperparameters, training schedule) are deferred to the Appendices.

3.1 Problem Formulation

We are given an observed motion sequence $M_{1:t} = \{x_1, \ldots, x_t\}$, where each frame $x_\tau = (\boldsymbol{r}_\tau, \boldsymbol{\theta}_\tau)$ consists of root motion $\boldsymbol{r}_\tau = (\boldsymbol{t}_\tau, \boldsymbol{R}_\tau)$, with translation $\boldsymbol{t}_\tau \in \mathbb{R}^3$ and orientation $\boldsymbol{R}_\tau \in SO(3)$,

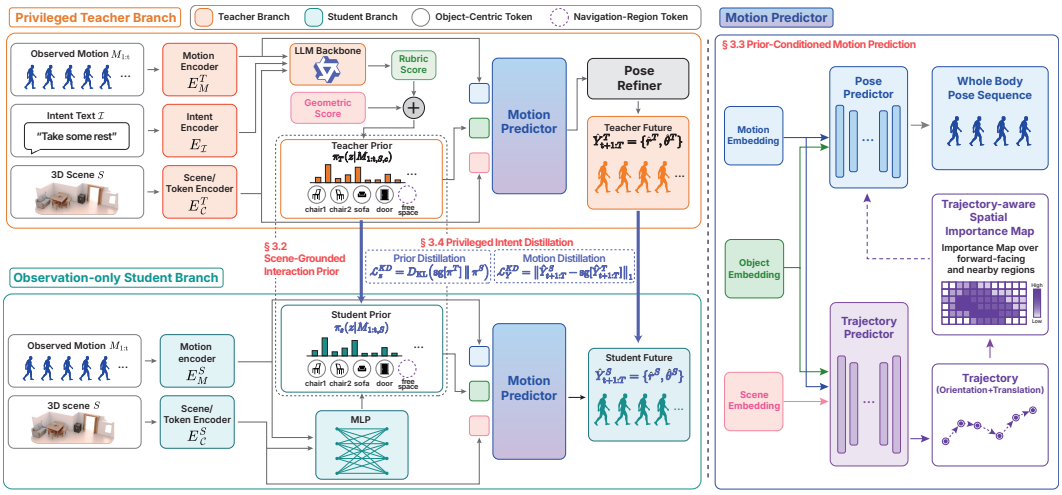


Figure 2: **Overall pipeline of *SGIP*.** A privileged teacher branch (top, training only) constructs an intent-conditioned prior $\pi^{\mathcal{T}}$ over candidate grounding tokens by fusing an LLM-based rubric score with a geometric compatibility score, with a Pose Refiner further intent-aligning the teacher's pose. An observation-only student branch (bottom) predicts $\pi^{\mathcal{S}}$ from motion $M_{1:t}$ and scene S alone via a scoring head $F_{\mathcal{S}}^{\text{grd}}$ (a permutation-equivariant MLP), trained via prior distillation $\mathcal{L}_z^{KD}$ and motion-level distillation $\mathcal{L}_Y^{KD}$. The selected token then conditions the motion predictor (right) that outputs future trajectory and full-body pose.

together with SMPL-X body pose $\boldsymbol{\theta}_\tau$ [44]. The surrounding 3D environment is a point cloud $S \in \mathbb{R}^{N\times 3}$. Our goal is to predict the future motion $\hat{Y}_{t+1:T} = \{\hat{x}_{t+1}, \ldots, \hat{x}_T\}$ over a horizon $\Delta T = T - t$. To make scene-interaction reasoning explicit, we extract a set of candidate *grounding tokens* $\mathcal{C} = \mathcal{C}^{\text{obj}} \cup \mathcal{C}^{\text{nav}}$, where $\mathcal{C}^{\text{obj}}$ contains object instances, and $\mathcal{C}^{\text{nav}}$ is a single scene-global token representing object-agnostic navigable regions. Since multiple tokens in $\mathcal{C}$ can afford plausible futures from the same observed motion, we introduce a *scene-grounded interaction prior* π over $\mathcal{C}$:

$$\pi_i = p(z = i \mid M_{1:t}, S), \quad i \in \mathcal{C}, \tag{1}$$

where the latent z indexes the token on which future motion is grounded.

At training time, we obtain a textual intent $\mathcal{I}$ (e.g., *"take some rest"*) by applying a frozen vision-language model to training videos and corresponding 3D motion visualizations (Appendix C). The intent and an LLM-based rubric scorer construct a privileged teacher prior $\pi^{\mathcal{T}}$, which is distilled into the observation-only student prior $\pi^{\mathcal{S}}$. The most probable token under $\pi^{\mathcal{S}}$ conditions the motion predictor, augmented at training with prior- and motion-level distillation losses against the teacher. At inference, $\mathcal{I}$ and the LLM are discarded.

3.2 Scene-Grounded Interaction Prior

We model future-motion grounding as a categorical distribution π over the candidate set $\mathcal{C}$, and construct two views of this distribution: a privileged *teacher* prior $\pi^{\mathcal{T}}$ that exploits intent and an LLM at training time, and an observation-only *student* prior $\pi^{\mathcal{S}}$ that approximates $\pi^{\mathcal{T}}$ from motion and scene alone at inference.

Token representations. Object instances $\mathcal{C}^{\text{obj}}$ are extracted via zero-shot 3D instance segmentation [26] on S. Each token $c_i = (\boldsymbol{\phi}_i, \boldsymbol{e}_i, \boldsymbol{m}_i)$ consists of three representations that capture complementary aspects of the token:

$$\boldsymbol{\phi}_i = [\boldsymbol{\phi}_{\text{geo}}(c_i),\ \boldsymbol{\phi}_{\text{rel}}(c_i, M_{1:t})] \qquad \text{(geometric descriptor)}, \tag{2}$$

$$\boldsymbol{e}_i = E_{\mathcal{C}}(c_i) \qquad \text{(token embedding)}, \tag{3}$$

$$\boldsymbol{m}_i \subseteq S \qquad \text{(point-cloud mask)}. \tag{4}$$

The $\boldsymbol{\phi}_i$ captures both where the token is in space and how the observed motion is approaching it, enabling reasoning about which specific instance among same-class candidates is being headed

toward. The $\boldsymbol{e}_i$ is a learned representation of the token's semantic appearance, capturing the kind of interaction it affords beyond raw geometry. The $\boldsymbol{m}_i$ provides the token's 3D shape in S, grounding reasoning about physically plausible body configurations at the interaction target. The observed motion sequence is encoded into a single embedding $h_M^b = E_M^b(M_{1:t}) \in \mathbb{R}^{d_M}$ for branch $b \in \{\mathcal{T}, \mathcal{S}\}$, where E_M^b denotes the branch's motion encoder.

Teacher prior: combining semantic and geometric evidence. At training, the teacher constructs $\pi^{\mathcal{T}}$ by combining two complementary scores. A *semantic relevance score* obtained from an LLM rates each candidate against the textual intent $\mathcal{I}$:

$$s_i^{\text{rubric}} = \text{LLM}(\mathcal{I}, c_i), \quad i \in \mathcal{C}, \tag{5}$$

where the LLM is queried with a fixed prompt scoring two criteria—intent affordance (how well a candidate semantically supports the intent) and intent usability (how easily the intent can be realized using that candidate, given the scene). The full prompt and rubric are in Appendix B (see also Appendix C.1).

A *geometric compatibility score* aggregates fixed directional alignments between the observed motion and each candidate (Details in Appendix D.4).

$$s_i^{\text{geo}} = w_{\text{face}}\, \text{align}_{\text{face}}(\phi_i, M_{1:t}) + w_{\text{body}}\, \text{align}_{\text{body}}(\phi_i, M_{1:t}) + w_{\text{line}}\, \text{align}_{\text{line}}(\phi_i, M_{1:t}). \tag{6}$$

The two scores are complementary. The LLM captures intent–class affordance (*a chair affords sitting*) but cannot disambiguate among multiple instances of the same class within a scene; the geometric score grounds the semantic prior in the observed motion, identifying *which* specific instance is being approached. We fuse them via temperature softmax with fusion weight α and temperature τ:

$$\pi_i^{\mathcal{T}} = \frac{\exp\big((s_i^{\text{rubric}} + \alpha\, s_i^{\text{geo}})/\tau\big)}{\sum_{j \in \mathcal{C}} \exp\big((s_j^{\text{rubric}} + \alpha\, s_j^{\text{geo}})/\tau\big)}. \tag{7}$$

This yields an intent-conditioned prior that the student is trained to match.

Student prior: observation-only approximation. At inference, neither $\mathcal{I}$ nor the LLM is available, so the student reproduces π from motion and the geometric token descriptors alone through a learnable scoring head $F_{\mathcal{S}}^{\text{grd}}$:

$$\pi_i^{\mathcal{S}} = \frac{\exp F_{\mathcal{S}}^{\text{grd}}(h_M^{\mathcal{S}}, \phi_i)}{\sum_{j \in \mathcal{C}} \exp F_{\mathcal{S}}^{\text{grd}}(h_M^{\mathcal{S}}, \phi_j)}. \tag{8}$$

Because $\mathcal{C}$ varies in size across scenes and tokens have no canonical order, $F_{\mathcal{S}}^{\text{grd}}$ is implemented as a permutation-equivariant MLP over candidates [65]. The student is trained to match $\pi^{\mathcal{T}}$ via knowledge distillation (Section 3.4), inheriting the teacher's intent-aware grounding without any auxiliary signal at inference.

3.3 Prior-Conditioned Motion Prediction

Given the prior π^b from branch $b \in \{\mathcal{T}, \mathcal{S}\}$, the motion predictor D^b produces future motion $\hat{Y}_{t+1:T}^b = \{\hat{\boldsymbol{r}}_{t+1:T}^b, \hat{\boldsymbol{\theta}}_{t+1:T}^b\}$ conditioned on the most probable grounding token,

$$i^* = \arg\max_{i \in \mathcal{C}} \pi_i^b, \tag{9}$$

whose embedding $\boldsymbol{e}_{i^*}$ and point-cloud mask $\boldsymbol{m}_{i^*}$ are read off from the selected token c_{i^*}. For multi-future prediction, we separately decode the top-K tokens ranked by $\pi^{\mathcal{S}}$ while keeping the observed motion, scene, and model parameters fixed. Only the selected token's embedding and mask change, yielding K token-conditioned future motions without stochastic sampling or privileged inputs. As shown on the right side of Fig. 2, D^b follows a two-stage trajectory-then-pose paradigm [38].

Trajectory branch. The trajectory branch first encodes the 3D scene S into per-point features via a scene encoder E_S, then cross-attends the motion embedding $\boldsymbol{h}_{\boldsymbol{M}}^{\boldsymbol{b}}$ to these features (both encoders illustrated in Figure 2), with the selected token's embedding $\boldsymbol{e}_{i^*}$ injected as a bias on the attention logits to focus attention on regions around the chosen interaction target:

$$\hat{r}_{t+1:T}^b = D_{\text{traj}}^b(h_M^b, S, e_{i^*}). \tag{10}$$

The predicted trajectory then induces a spatial importance map P^b aggregating path proximity, forward-facing alignment, and proximity to the selected token's region:

$$P^b = f_{\text{spatial}}(\hat{r}^b_{t+1:T}, S, m_{i^*}). \tag{11}$$

Pose branch. P^b is added as a bias to the pose branch's scene-attention logits, keeping body pose consistent with the planned path and interaction target:

$$\tilde{\theta}^b_{t+1:T} = D^b_{\text{pose}}(h^b_M, \hat{r}^b_{t+1:T}, S, P^b). \tag{12}$$

Architectural details (f_{spatial} form, encoders, pose refiner) are in Appendix D. For the teacher only, a pose refiner Refine that leverages $\mathcal{I}$ is applied to recover intent-conditioned pose semantics beyond motion reconstruction (e.g., distinguishing sitting from standing in front of the same chair); details are in Appendix D.7.

3.4 Privileged Intent Distillation

To deploy the teacher's intent-aware reasoning at inference, we distill it into the student via two complementary signals: *prior distillation* aligns *where* future motion is grounded ($\pi^{\mathcal{T}} \to \pi^{\mathcal{S}}$), and *motion-level distillation* aligns *how* the model moves toward that target ($\hat{Y}^{\mathcal{T}} \to \hat{Y}^{\mathcal{S}}$).

Prior distillation. We align $\pi^{\mathcal{S}}$ with $\pi^{\mathcal{T}}$ via forward KL divergence:

$$\mathcal{L}^{\text{KD}}_z = D_{\text{KL}}\big(\text{sg}[\pi^{\mathcal{T}}] \,\|\, \pi^{\mathcal{S}}\big) = \sum_{i \in \mathcal{C}} \pi^{\mathcal{T}}_i \log \frac{\pi^{\mathcal{T}}_i}{\pi^{\mathcal{S}}_i}, \tag{13}$$

where $\text{sg}[\cdot]$ stops gradients to $\pi^{\mathcal{T}}$. The mass-covering forward direction forces the student to cover every token the teacher considers, preserving the full intent-conditioned grounding structure rather than only its top-1 and preventing modal collapse.

Motion-level distillation. We distill the teacher's motion via L_1 loss on trajectory and pose:

$$\mathcal{L}^{\text{KD}}_Y = \lambda_{\text{traj}} \big\| \hat{r}^{\mathcal{S}}_{t+1:T} - \text{sg}[\hat{r}^{\mathcal{T}}_{t+1:T}] \big\|_1 + \lambda_{\text{pose}} \big\| \tilde{\theta}^{\mathcal{S}}_{t+1:T} - \text{sg}[\hat{\theta}^{\mathcal{T}}_{t+1:T}] \big\|_1, \tag{14}$$

where $\hat{\boldsymbol{\theta}}^{\mathcal{T}}_{t+1:T}$ is the refined teacher pose, and $\lambda_{\text{traj}}, \lambda_{\text{pose}}$ balance the two terms. Crucially, since the teacher pose is refined with intent, the intent-conditioned pose semantics (e.g., sitting versus standing in front of the same chair) can be distilled into the student.

Training schedule. The teacher is trained first with its motion loss $\mathcal{L}^{\mathcal{T}}_{\text{mot}}$ and additional Refine-specific terms (Appendix D.8). The student then freezes the entire teacher and optimizes

$$\mathcal{L}^{\mathcal{S}} = \mathcal{L}^{\mathcal{S}}_{\text{mot}} + \lambda_z \, \mathcal{L}^{\text{KD}}_z + \lambda_Y \, \mathcal{L}^{\text{KD}}_Y, \tag{15}$$

where $\mathcal{L}^{\mathcal{S}}_{\text{mot}}$ shares the form of $\mathcal{L}^{\mathcal{T}}_{\text{mot}}$.

4 Experiments

4.1 Experimental Setup

Dataset. We evaluate *SGIP* on the GIMO benchmark [69], a real-world indoor dataset providing full-body SMPL-X motion, reconstructed 3D scenes, and per-sequence textual intent annotations. Following the standard scene-aware HMP protocol [38, 63, 69], the model observes a motion prefix in a 3D scene and predicts the future global trajectory and full-body pose. We additionally evaluate on PROX [17], which provides reconstructed 3D scenes and SMPL-X motion but no gaze or intent annotations; gaze-using baselines are therefore evaluated in their gaze-free variants. The full SGIP pipeline is trained and evaluated on PROX under the same protocol as on GIMO.

Baselines. We compare against two groups. *Pose-only HMP* baselines—LTD [39], SPGSN [35], and AuxFormer [59]—forecast motion from body dynamics alone. *Scene- and gaze-aware* baselines—BiFu [69], SIF3D [38], and GAP3DS [63]—condition on 3D scene context with gaze as an auxiliary

Table 1: Main comparison on GIMO and PROX. The "Auxiliary Cue" column denotes auxiliary signals beyond $M + S$: g=gaze, $\mathcal{I}$=textual intent. †: diagnostic privileged branch on GIMO (not deployed); $*$: our gaze-free re-evaluation of gaze-using baselines. SGIP results on GIMO are mean±std over 5 seeds. **Bold**/underline: best/second-best within block.

	Method	Auxiliary Cue	Traj-path ↓	Traj-dest ↓	MPJPE-path ↓	MPJPE-dest ↓
GIMO	*Models with additional or privileged cues*					
	LTD [39] w/ gaze	g	655	801	155.6	202.0
	SPGSN [35] w/ gaze	g	702	982	156.0	202.0
	AuxFormer [59] w/ gaze	g	622	746	161.5	201.1
	BiFu [69]	g	661	727	167.8	205.0
	SIF3D [38]	g	590	666	156.6	195.7
	GAP3DS [63]	g	575	<u>623</u>	**141.2**	**171.4**
	SGIP$^{\dagger}_{\text{priv}}$	g	<u>528</u>	635	<u>149.9</u>	190.6
	SGIP$^{\dagger}_{\text{priv}}$	$\mathcal{I}$	**485**	**556**	151.2	<u>187.8</u>
	Observation-only models					
	LTD [39]	–	673	890	<u>156.4</u>	203.0
	SPGSN [35]	–	700	893	157.0	202.2
	AuxFormer [59]	–	732	902	165.1	202.4
	SIF3D* [38]	–	700	823	164.2	<u>199.6</u>
	GAP3DS* [63]	–	<u>642</u>	<u>701</u>	170.1	206.4
	SGIP	–	**549** (±5.0)	**681** (±7.6)	**153.1** (±1.8)	**195.1** (±2.6)
PROX	LTD [39]	–	677	753	223.4	226.6
	SPGSN [35]	–	595	663	286.3	271.3
	AuxFormer [59]	–	553	727	<u>199.2</u>	240.8
	SIF3D* [38]	–	<u>488</u>	<u>659</u>	228.5	<u>217.4</u>
	SGIP	–	**470**	**576**	**194.6**	**197.7**

cue. To match our deployment assumption, we additionally evaluate gaze-free variants of SIF3D and GAP3DS (denoted *). We report the deployed student as *SGIP* and the privileged teacher as *SGIP*$^{\dagger}_{\text{priv}}$ (diagnostic upper bound only).

Metrics. Following prior work [38, 63, 69], we report trajectory and pose accuracy. *Traj-path*/*Traj-dest* measure global trajectory error averaged over the prediction horizon and at the final frame, respectively. *MPJPE-path*/*MPJPE-dest* report mean per-joint position error [27, 39] on the same two horizons.

Implementation. Object-centric tokens are obtained via OpenIns3D [26]; semantic rubric scores use Qwen3-14B [61], prompted once per (sequence, token) pair and cached offline. Full architectural and training details are in Appendix D.5 and F.

4.2 Main Results

Table 1 demonstrates *SGIP*'s consistent advantage over scene-aware baselines on both GIMO and PROX. Pose-only HMP methods exhibit large trajectory errors, confirming that 3D scene context is necessary for long-horizon grounding. Scene-aware baselines improve over pose-only counterparts but still trail *SGIP*: scene-level conditioning alone does not resolve *which* object the future motion engages. Gaze-using methods narrow this gap, yet remain limited because gaze encodes *where the subject looks* rather than *what the subject intends to do*.

SGIP's performance gains stem from three design factors. **(1) Explicit scene-grounded interaction prior.** The predicted prior π^S selects a discrete grounding token rather than relying on implicit attention, providing both a spatial mask and a semantic embedding that anchor the predictor to a specific interaction target. This contributes to the largest gains on trajectory metrics, where grounding ambiguity dominates. **(2) Privileged intent distillation.** The teacher trained with textual intent ($\mathcal{I}$) transfers an intent-aware grounding distribution and motion structure to the student, enabling inference from motion and scene alone. The privileged teacher diagnostic (*SGIP*$^{\dagger}_{\text{priv}}$) quantifies the headroom available from explicit intent and serves as a supervision upper bound, not a deployable model. **(3) Cross-dataset generalization.** The same architecture trained from scratch on PROX maintains the trend, outperforming the strongest scene-aware baseline SIF3D* across all four met-

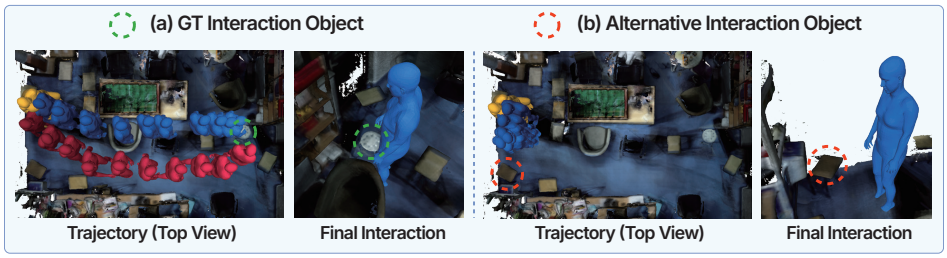


Figure 3: **Multimodal futures from π^S.** We condition the same fixed student predictor D^S on two different grounding tokens drawn from π^S: **(a)** the ground-truth interaction object (green dashed circle), and **(b)** an alternative high-probability token of π^S (red dashed circle). Each panel visualizes the resulting predicted trajectory (top-down view) and the final interaction pose; only the selected token $(\boldsymbol{m}_i, \boldsymbol{e}_i)$ fed into D^S varies.

Table 2: Ablation studies on GIMO. **(a)** Contribution of distillation components $\mathcal{L}^z_{\text{KD}}$ (prior) and $\mathcal{L}^Y_{\text{KD}}$ (motion). **(b)** Hard vs. soft top-k aggregation of grounding tokens at inference. **(c)** Top-k accuracy of ranking the GIMO ground-truth interaction token at position k. Variants in (a)–(b) are observation-only; in (c), π^T serves as a privileged upper bound.

(a) Distillation components.

KD		Traj ↓		MPJPE ↓	
z	Y	path	dest	path	dest
		740	838	169.1	204.8
	✓	856	896	164.1	198.9
✓		574	688	**151.8**	198.6
✓	✓	**549**	**681**	153.1	**195.1**

(b) Hard vs. soft top-k.

Aggregation	Traj ↓		MPJPE ↓	
	path	dest	path	dest
Hard (top-1)	**549**	**681**	**153.1**	**195.1**
Soft (top-2)	584	736	158.3	210.3
Soft (top-3)	715	877	176.5	210.2

(c) Cumulative Grounding acc (%).

Method	Top-1	Top-2	Top-3
Rubric-only	54	24	2
Geometric-only	51	16	11
Teacher π^T	**86**	5	3
Student π^S	76	5	3

rics with particularly large margins on pose accuracy and trajectory destination, indicating that the benefits do not arise from GIMO-specific tuning.

These results support the central hypothesis of this work: explicit scene-grounded interaction priors, distilled from privileged intent, capture structure that gaze and scene-attention baselines do not.

4.3 Ablation Studies

Table 2a dissects the student objective $\mathcal{L}^S$. Without any distillation, the student performs poorly on trajectory, confirming that 3D scene conditioning alone does not resolve grounding ambiguity. Prior distillation $\mathcal{L}^z_{\text{KD}}$ alone yields the largest single-component gain, with the distilled grounding distribution steering the student toward the correct interaction target and anchoring both trajectory and pose prediction.

Applying $\mathcal{L}^Y_{\text{KD}}$ in isolation, without prior distillation, *degrades* trajectory accuracy below the no-distillation baseline. We attribute this to *grounding misalignment*: the teacher's output is conditioned on its selected token i^*_T, while the student without prior supervision selects an inconsistent token i^*_S, so imitating the teacher's trajectory and pose under a different latent grounding pulls predictions toward geometrically incompatible targets. This supports our framework choice that prior distillation is a prerequisite for, not a parallel signal to, output-level distillation. Combining both signals attains the best overall performance, indicating that once grounding is correctly transferred, mimicking the teacher's intent-aware motion structure provides further non-trivial gains.

Hard vs. soft top-k aggregation. We compare hard top-1 token selection (Eq. 9) against soft top-k aggregation, where the predictor consumes a probability-weighted mixture of multiple tokens' masks and embeddings (Appendix G.2). As shown in Table 2b, hard top-1 consistently outperforms soft top-2 and top-3 on all four metrics, validating our discrete grounding-selector design for single-future prediction.

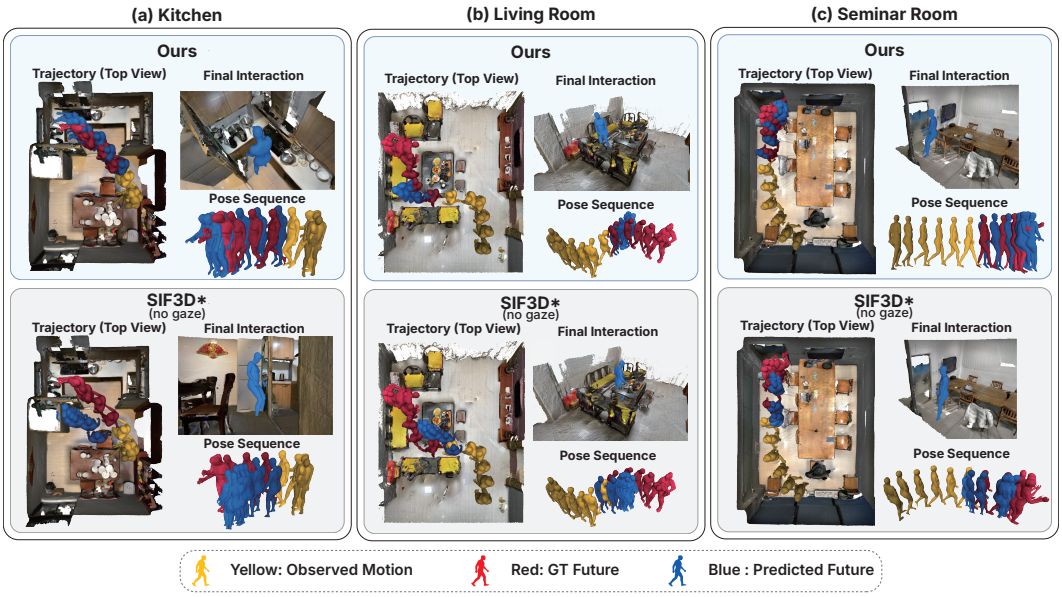


Figure 4: Qualitative comparison on three GIMO scenes (Kitchen, Living Room, Seminar Room). For each scene, the top row shows *SGIP* (ours) and the bottom row shows the gaze-free baseline SIF3D*. Each row visualizes the predicted trajectory (top-down view), the final interaction frame, and the full-body pose sequence. Yellow: observed motion; red: GT future; blue: predicted future.

Multimodality of the learned prior. While hard top-1 is optimal for single-future prediction, we examine whether π^S itself encodes meaningful structure beyond the argmax. Conditioning D^S on each of the top-3 grounding tokens of π^S in turn (Figure 3), we find that top-2 and top-3 candidates correspond to semantically distinct interactions (e.g., a different chair instance or a different object category), each producing scene-consistent trajectory and pose rather than degenerate variants of the top-1. This indicates that π^S preserves genuine multimodal structure and that D^S has learned a token-conditional motion distribution—a property our framework retains for the diverse-future extension.

Grounding token selection. Table 2c reports top-1 accuracy on the GIMO-annotated interaction target. The teacher π^T achieves 86% versus 54%/51% for rubric-/geometric-only baselines, confirming that semantic intent and motion-grounded geometry are complementary signals. The deployed student π^S retains 76%, directly evidencing that distillation transfers intent-aware grounding to an observation-only model. See Appendix G.1 for full top-k analysis and candidate-set statistics.

4.4 Qualitative Results

Figure 4 compares *SGIP* with SIF3D* on three cluttered GIMO scenes. Across all three, *SGIP* more closely tracks the ground-truth trajectory shape and final body placement than the gaze-free baseline. The gap is most pronounced when multiple scene regions could plausibly attract the future motion, illustrating the role of the learned scene-grounded interaction prior in disambiguating competing futures—a structural advantage that scene-attention baselines lacking explicit grounding cannot match. Appendix H provides two further analyses that complement these examples: a side-by-side comparison of SIF3D (with/without gaze) and the SGIP teacher/student, showing that the deployable student closely tracks the privileged teacher's correct grounding while SIF3D drifts in both gaze configurations, and a top-k visualization confirming that π^S encodes semantically distinct, scene-consistent alternative futures. Additional qualitative video results are available in the supplementary material.

5 Conclusion and Limitation

We presented *SGIP*, which learns a scene-grounded interaction prior for scene-aware human motion prediction by distilling intent-aware supervision from a privileged teacher into an observation-only student. At training, the teacher leverages textual intent and an LLM-based rubric scorer to construct an interaction distribution over object-centric and navigation tokens; the student approximates this distribution from motion and scene alone, and conditions trajectory and full-body pose prediction on the selected grounding. On both GIMO and PROX, SGIP consistently improves trajectory and pose accuracy over scene-aware observation-only baselines, with the privileged teacher serving as an empirical upper bound on intent-aware reasoning. SGIP nevertheless relies on the quality of candidate grounding tokens—degrading under noisy scene geometry or incomplete zero-shot segmentation—and assumes textual intent as privileged supervision during training; extending the framework to settings without such annotations (e.g., by mining pseudo-intent from unlabeled human–scene trajectories) and to diverse multi-future generation by retaining the full predicted prior are natural future directions.

References

[1] Michael Ahn, Anthony Brohan, Noah Brown, Yevgen Chebotar, Omar Cortes, Byron David, Chelsea Finn, Chuyuan Fu, Keerthana Gopalakrishnan, Karol Hausman, et al. Do as i can, not as i say: Grounding language in robotic affordances. In *Proceedings of the Conference on Robot Learning (CoRL)*, 2022.

[2] Emre Aksan, Manuel Kaufmann, Peng Cao, and Otmar Hilliges. A spatio-temporal transformer for 3D human motion prediction. In *International Conference on 3D Vision (3DV)*, 2021.

[3] Alexandre Alahi, Kratarth Goel, Vignesh Ramanathan, Alexandre Robicquet, Li Fei-Fei, and Silvio Savarese. Social LSTM: Human trajectory prediction in crowded spaces. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016.

[4] Sadegh Aliakbarian, Fatemeh Sadat Saleh, Mathieu Salzmann, Lars Petersson, and Stephen Gould. A stochastic conditioning scheme for diverse human motion prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5223–5232, 2020.

[5] German Barquero, Sergio Escalera, and Cristina Palmero. BeLFusion: Latent diffusion for behavior-driven human motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2317–2327, 2023.

[6] Zhe Cao, Hang Gao, Karttikeya Mangalam, Qi-Zhi Cai, Minh Vo, and Jitendra Malik. Long-term human motion prediction with scene context. In *European Conference on Computer Vision (ECCV)*, 2020.

[7] Zhi Cen, Huaijin Pi, Sida Peng, Zehong Shen, Minghui Yang, Shuai Zhu, Hujun Bao, and Xiaowei Zhou. Generating human motion in 3D scenes from text descriptions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1855–1866, 2024.

[8] Ling-Hao Chen, Jiawei Zhang, Yewen Li, Yiren Pang, Xiaobo Xia, and Tongliang Liu. HumanMAC: Masked motion completion for human motion prediction. In *Proceedings of the IEEE/CVF international conference on computer vision*, pages 9544–9555, 2023.

[9] Enric Corona, Albert Pumarola, Guillem Alenya, and Francesc Moreno-Noguer. Context-aware human motion prediction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6992–7001, 2020.

[10] Lingwei Dang, Yongwei Nie, Chengjiang Long, Qing Zhang, and Guiqing Li. MSR-GCN: Multi-scale residual graph convolution networks for human motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11467–11476, 2021.

[11] Markos Diomataris, Nikos Athanasiou, Omid Taheri, Xi Wang, Otmar Hilliges, and Michael J. Black. WANDR: Intention-guided human motion generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 927–936, 2024.

[12] Katerina Fragkiadaki, Sergey Levine, Panna Felsen, and Jitendra Malik. Recurrent network models for human dynamics. In *IEEE International Conference on Computer Vision (ICCV)*, 2015.

[13] Xuehao Gao, Yang Yang, Yang Wu, Shaoyi Du, and Guo-Jun Qi. Multi-condition latent diffusion network for scene-aware neural human motion prediction. *IEEE transactions on Image Processing (TIP)*, pages 3907–3920, 2024.

[14] Nuno C. Garcia, Pietro Morerio, and Vittorio Murino. Modality distillation with multiple stream networks for action recognition. In *European Conference on Computer Vision (ECCV)*, 2018.

[15] Anindita Ghosh, Rishabh Dabral, Vladislav Golyanik, Christian Theobalt, and Philipp Slusallek. IMoS: Intent-driven full-body motion synthesis for human-object interactions. *Computer Graphics Forum*, 42(2):1–12, 2023.

[16] Liang-Yan Gui, Kevin Zhang, Yu-Xiong Wang, Xiaodan Liang, José MF Moura, and Manuela Veloso. Teaching robots to predict human motion. In *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 562–567. IEEE, 2018.

[17] Mohamed Hassan, Vasileios Choutas, Dimitrios Tzionas, and Michael J. Black. Resolving 3D human pose ambiguities with 3D scene constraints. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 2282–2292, 2019.

[18] Mohamed Hassan, Duygu Ceylan, Ruben Villegas, Jun Saito, Jimei Yang, Yi Zhou, and Michael J Black. Stochastic scene-aware motion prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[19] Mohamed Hassan, Partha Ghosh, Joachim Tesch, Dimitrios Tzionas, and Michael J Black. Populating 3d scenes by learning human-scene interaction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 14708–14718, 2021.

[20] Mary Hayhoe and Dana Ballard. Eye movements in natural behavior. *Trends in cognitive sciences*, 9(4):188–194, 2005.

[21] Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.

[22] Judy Hoffman, Saurabh Gupta, and Trevor Darrell. Learning with side information through modality hallucination. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 826–834, 2016.

[23] Yining Hong, Haoyu Zhen, Peihao Chen, Shuhong Zheng, Yilun Du, Zhenfang Chen, and Chuang Gan. 3D-LLM: Injecting the 3D world into large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[24] Jiangyong Huang, Silong Yong, Xiaojian Ma, Xiongkun Linghu, Puhao Li, Yan Wang, Qing Li, Song-Chun Zhu, Baoxiong Jia, and Siyuan Huang. An embodied generalist agent in 3d world. *arXiv preprint arXiv:2311.12871*, 2023.

[25] Siyuan Huang, Zan Wang, Puhao Li, Baoxiong Jia, Tengyu Liu, Yixin Zhu, Wei Liang, and Song-Chun Zhu. Diffusion-based generation, optimization, and planning in 3D scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 16750–16761, 2023.

[26] Zhening Huang, Xiaoyang Wu, Xi Chen, Hengshuang Zhao, Lei Zhu, and Joan Lasenby. Openins3d: Snap and lookup for 3d open-vocabulary instance segmentation. In *European Conference on Computer Vision (ECCV)*, 2024.

[27] Catalin Ionescu, Dragos Papava, Vlad Olaru, and Cristian Sminchisescu. Human3.6M: Large scale datasets and predictive methods for 3D human sensing in natural environments. *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 36(7), 2014.

[28] Ashesh Jain, Amir R. Zamir, Silvio Savarese, and Ashutosh Saxena. Structural-RNN: Deep learning on spatio-temporal graphs. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5308–5317, 2016.

[29] Jaewoo Jeong, Seohee Lee, Daehee Park, Giwon Lee, and Kuk-Jin Yoon. Multi-modal knowledge distillation-based human trajectory forecasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 24222–24233, 2025.

[30] Biao Jiang, Xin Chen, Wen Liu, Jingyi Yu, Gang Yu, and Tao Chen. MotionGPT: Human motion as a foreign language. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[31] Nan Jiang, Zhiyuan Zhang, Hongjie Li, Xiaoxuan Ma, Zan Wang, Yixin Chen, Tengyu Liu, Yixin Zhu, and Siyuan Huang. Scaling up dynamic human-scene interaction modeling. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 1737–1747, 2024.

[32] Nilesh Kulkarni, Davis Rempe, Kyle Genova, Abhijit Kundu, Justin Johnson, David Fouhey, and Leonidas Guibas. NIFTY: Neural object interaction fields for guided human motion synthesis. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

[33] Michael F Land and Mary Hayhoe. In what ways do eye movements contribute to everyday activities? *Vision research*, 41(25-26):3559–3565, 2001.

[34] Jiaman Li, Alexander Clegg, Roozbeh Mottaghi, Jiajun Wu, Xavier Puig, and C. Karen Liu. Controllable human-object interaction synthesis. In *European Conference on Computer Vision (ECCV)*, 2024.

[35] Maosen Li, Siheng Chen, Zijing Zhang, Lingxi Xie, Qi Tian, and Ya Zhang. Skeleton-parted graph scattering networks for 3D human motion prediction. In *European Conference on Computer Vision (ECCV)*, 2022.

[36] Yi Lin, Wangjie Xia, and Ting Yu. HuMo3D: Intention-driven dual-branch multimodal human motion prediction in 3D scenes. *IEEE Signal Processing Letters*, 33:614–618, 2026. doi: 10.1109/LSP.2026.3652960.

[37] David Lopez-Paz, Léon Bottou, Bernhard Schölkopf, and Vladimir Vapnik. Unifying distillation and privileged information. *arXiv preprint arXiv:1511.03643*, 2015.

[38] Zhenyu Lou, Qiongjie Cui, Haofan Wang, Xu Tang, and Hong Zhou. Multimodal sense-informed forecasting of 3D human motions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2144–2154, 2024.

[39] Wei Mao, Miaomiao Liu, Mathieu Salzmann, and Hongdong Li. Learning trajectory dependencies for human motion prediction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019.

[40] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. History repeats itself: Human motion prediction via motion attention. In *Proceedings of the European Conference on Computer Vision (ECCV)*, 2020.

[41] Wei Mao, Miaomiao Liu, and Mathieu Salzmann. Generating smooth pose sequences for diverse human motion prediction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2021.

[42] Wei Mao, Miaomiao Liu, Richard I. Hartley, and Mathieu Salzmann. Contact-aware human motion forecasting. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 7356–7367, 2022.

[43] Julieta Martinez, Michael J. Black, and Javier Romero. On human motion prediction using recurrent neural networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[44] Georgios Pavlakos, Vasileios Choutas, Nima Ghorbani, Timo Bolkart, Ahmed A. A. Osman, Dimitrios Tzionas, and Michael J. Black. Expressive body capture: 3D hands, face, and body from a single image. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 10975–10985, 2019.

[45] Charles R. Qi, Hao Su, Kaichun Mo, and Leonidas J. Guibas. PointNet: Deep learning on point sets for 3D classification and segmentation. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.

[46] Charles R. Qi, Li Yi, Hao Su, and Leonidas J. Guibas. PointNet++: Deep hierarchical feature learning on point sets in a metric space. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[47] Adriana Romero, Nicolas Ballas, Samira Ebrahimi Kahou, Antoine Chassang, Carlo Gatta, and Yoshua Bengio. FitNets: Hints for thin deep nets. In *International Conference on Learning Representations (ICLR)*, 2015.

[48] Tim Salzmann, Boris Ivanovic, Punarjay Chakravarty, and Marco Pavone. Trajectron++: Dynamically-feasible trajectory forecasting with heterogeneous data. In *European Conference on Computer Vision (ECCV)*, 2020.

[49] Tim Salzmann, Marco Pavone, and Markus Ryll. Motron: Multimodal probabilistic human motion forecasting. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6457–6466, 2022.

[50] Theodoros Sofianos, Alessio Sampieri, Luca Franco, and Fabio Galasso. Space-time-separable graph convolutional network for pose forecasting. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 11209–11218, 2021.

[51] Caiyi Sun, Yujing Sun, Xiao Han, Zemin Yang, Jiawei Liu, Xinge Zhu, Siu Ming Yiu, and Yuexin Ma. Humof: Human motion forecasting in interactive social scenes. In *International Conference on Learning Representations*, volume 2026, pages 6511–6534, 2026.

[52] Vladimir Vapnik and Akshay Vashist. A new learning paradigm: Learning using privileged information. *Neural Networks*, 22(5–6):544–557, 2009.

[53] Jiashun Wang, Huazhe Xu, Jingwei Xu, Sifei Liu, and Xiaolong Wang. Synthesizing long-term 3D human motion and interaction in 3D scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9401–9411, 2021.

[54] Tuo Wang, Zhenbo Song, Zhenyu Lou, Qiongjie Cui, Luoming Zhang, Cheng Cheng, Haofan Wang, Xu Tang, Huaxia Li, and Hong Zhou. Harmonizing stochasticity and determinism: Scene-responsive diverse human motion prediction. volume 37, pages 39784–39811, 2024.

[55] Wenjia Wang, Liang Pan, Zhiyang Dou, Jidong Mei, Zhouyingcheng Liao, Yuke Lou, Yifan Wu, Lei Yang, Jingbo Wang, and Taku Komura. Sims: Simulating stylized human-scene interactions with retrieval-augmented script generation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 14117–14127, October 2025.

[56] Zan Wang, Yixin Chen, Tengyu Liu, Yixin Zhu, Wei Liang, and Siyuan Huang. HUMANISE: Language-conditioned human motion generation in 3D scenes. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, 2022.

[57] Zan Wang, Yixin Chen, Baoxiong Jia, Puhao Li, Jinlu Zhang, Jingze Zhang, Tengyu Liu, Yixin Zhu, Wei Liang, and Siyuan Huang. Move as you say, interact as you can: Language-guided human motion generation with scene affordance. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 433–444, 2024.

[58] Chaoyue Xing, Wei Mao, and Miaomiao Liu. Scene-aware human motion forecasting via mutual distance prediction. In *European Conference on Computer Vision (ECCV)*, 2024.

[59] Chenxin Xu, Robby T. Tan, Yuhong Tan, Siheng Chen, Yu Guang Wang, Xinchao Wang, and Yanfeng Wang. Auxiliary tasks benefit 3D skeleton-based human motion prediction. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

[60] Chenxin Xu, Robby T Tan, Yuhong Tan, Siheng Chen, Yu Guang Wang, Xinchao Wang, and Yanfeng Wang. EqMotion: Equivariant multi-agent motion prediction with invariant interaction reasoning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1410–1420, 2023.

[61] An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, et al. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.

[62] Jianing Yang, Xuweiyi Chen, Shengyi Qian, Nikhil Madaan, Madhavan Iyengar, David F Fouhey, and Joyce Chai. Llm-grounder: Open-vocabulary 3d visual grounding with large language model as an agent. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 7694–7701. IEEE, 2024.

[63] Ting Yu, Yi Lin, Jun Yu, Zhenyu Lou, and Qiongjie Cui. Vision-guided action: Enhancing 3D human motion prediction with gaze-informed affordance in 3D scenes. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12335–12346, 2025.

[64] Ye Yuan and Kris Kitani. DLow: Diversifying latent flows for diverse human motion prediction. In *European Conference on Computer Vision (ECCV)*, 2020.

[65] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[66] Jiahe Zhao, Ruibing Hou, Zejie Tian, Hong Chang, and Shiguang Shan. His-gpt: Towards 3d human-in-scene multimodal understanding. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 4317–4327, October 2025.

[67] Peisen Zhao, Lingxi Xie, Ya Zhang, and Qi Tian. Progressive privileged knowledge distillation for online action detection. *Pattern Recognition*, 129, 2022.

[68] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging LLM-as-a-judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

[69] Yang Zheng, Yanchao Yang, Kaichun Mo, Jiaman Li, Tao Yu, Yebin Liu, C. Karen Liu, and Leonidas J. Guibas. Gimo: Gaze-informed human motion prediction in context. In *European Conference on Computer Vision (ECCV)*, 2022.

A Training and Implementation details

This appendix elaborates on the training and inference protocol of our framework. We describe the two-stage training schedule of the teacher–student branches, summarize the implementation hyperparameters used in all experiments, and detail the inference procedure of the deployed observation-only branch.

Training schedule. We first train the privileged branch using $\mathcal{L}^T$. We then freeze the privileged branch and train the observation-only branch using $\mathcal{L}^S$, treating both the privileged distribution and privileged motion prediction as stop-gradient targets. The final model used for evaluation is the observation-only branch.

Implementation details. Both the privileged (teacher) and observation-only (student) branches are optimized with AdamW, using an initial learning rate of 3×10^{-4} and a weight decay of 1×10^{-4}. The learning rate is decayed every epoch by an `ExponentialLR` scheduler with $\gamma = 0.98$. We use a batch size of 8 and train each branch for 100 epochs. The student branch follows the identical optimization configuration as the teacher. All experiments are conducted on a single NVIDIA RTX 3090 GPU.

Inference. At inference, textual intent and the privileged branch are not used. Given only $M_{1:t}$ and S, the observation-only branch predicts $p^S(z \mid M_{1:t}, S)$, selects a single grounding token i_S^* via the rule of Eqs. 8–9 of the main paper, and forecasts the future motion

$$\hat{Y}^S_{t+1:T} = D^S\big(h^M,\, m_{i_S^*},\, e_{i_S^*}\big)\,. \tag{16}$$

Thus, the deployed model requires only motion and scene observations, while still benefiting from intent-guided scene-interaction reasoning learned during training.

B LLM-Based Rubric Scorer

This appendix specifies how the privileged teacher computes the per-token semantic relevance score s_i^{rubric} in Eq. 5 of the main paper.

B.1 Information Availability per Stage

Table 3 summarizes, for each stage of SGIP, the inputs that may be observed. The privileged teacher and the LLM-based scorer are restricted to training time; only the student is deployed at inference.

Stage	**Available inputs**
Privileged teacher (training only)	$M_{1:t}$, S, $\{c_i\}$, $\{\phi_i, e_i, m_i\}$, intent $\mathcal{I}$
Observation-only student (train + inference)	$M_{1:t}$, S, $\{c_i\}$, $\{\phi_i, m_i\}$

Table 3: Information availability across SGIP stages.

The teacher does not observe the ground-truth grounding token at any stage; it observes only the candidate tokens, their semantic labels, the textual intent c, and geometric descriptors $\{\phi_i\}$. Distillation thus transfers *intent-aware semantic reasoning* over candidate properties, not GT label leakage.

B.2 Criterion Rubric

A reasoning LLM scores each token c_i on two integer rubrics, each with an explicit point allocation:

$$\begin{aligned} a_i &\in [1,5] \quad (\textit{intent affordance}\text{: how well does } c_i \text{ semantically support the target intent } \mathcal{I}?) \\ u_i &\in [1,5] \quad (\textit{intent usability}\text{: given the scene } S\text{, how easy is it to realize } \mathcal{I} \text{ using } c_i?) \end{aligned} \tag{17}$$

The total $s_i^{\text{rubric}} = a_i + u_i \in [2, 10]$ is fed into the temperature softmax of Eq. (7)in the main paper.

Geometric alignment is already accessible to the student through ϕ_i and h_M, so the teacher's distinctive value is intent-aware semantic judgment. We encourage the rubric to reflect this by construction.

B.3 LLM Call and Reproducibility

We use **Qwen3-14B** [61] with NF4 4-bit quantization. Decoding uses greedy sampling (temperature $= 0.0$, top-$p = 1.0$, max new tokens $= 4096$, fixed seed). The prompt enforces a two-stage output: an intermediate natural-language rationale block (used to encourage scoring consistency, not parsed as supervision) followed by a strict JSON block listing per-token integer scores and the recomputed total. Each (sequence, token-set) pair is queried once and cached offline, so no LLM call occurs during training updates or inference.

B.4 JSON Validation and Exclusion Handling

Each JSON block is parsed against a schema that validates per-criterion ranges and recomputes $s_i^{\text{rubric}} = a_i + u_i$ when the LLM-reported total disagrees with the per-criterion sum. Out-of-range scores, missing per-token entries, or duplicate identifiers trigger a single retry. Validation concerns JSON structure only, not duplicated rubric values or repeated object names—both of which are legitimate (e.g., two chair instances).

If parsing still fails after the retry, we exclude the entire sample from teacher distillation; we do not drop individual tokens, since this would change the candidate-set cardinality and could remove the GT grounding token, altering the distillation target itself.

B.5 Stability Check

We verify rubric stability on a held-out subset of $N = 20$ samples scored five times each, measuring top-1 agreement and Kendall rank correlation. The 4-bit quantization is verified by comparison against bf16 inference on the same subset; we observed no significant drop in top-1 agreement.

C Intent Extraction from VLM

C.1 Intent Taxonomy

To represent user action intents in ego-centric environments, we define a taxonomy consisting of 16 intent categories. Each intent is selected to cover actions that frequently occur in everyday indoor environments, and the categories are organized based on the patterns of interaction with furniture and objects. Specifically, the 16 intents are defined as follows: *sit*, *sleep*, *drink*, *eat*, *read*, *use*, *take*, *open*, *light*, *clean*, *dress*, *throw*, *cook*, *write*, *fix*, along with an *unknown* class that handles actions not clearly belonging to any of the above categories.

The design of this taxonomy is grounded in two principles. First, the categories are constructed such that each intent is mutually exclusive while collectively covering the major behavioral patterns of daily activities (i.e., collectively exhaustive). Second, by defining each intent based on unit actions that are visually distinguishable from an ego-centric viewpoint, we ensure that the VLM can perform stable classification using visual cues alone. Furthermore, by explicitly introducing the *unknown* class, we prevent the model from being forced to assign incorrect labels to sequences that are ambiguous or do not fit any predefined category.

C.2 VLM-based Intent Extraction

To automatically label the intent of each sequence, we employ the Qwen3-VL-8B model. Since it is often difficult to accurately infer a user's action intent from a single viewpoint alone, we adopt a dual-view input strategy that simultaneously provides two types of visual information.

The first input is an **ego-centric video**, captured from the user's first-person viewpoint. This video provides fine-grained visual cues about the objects the user is looking at, hand movements, and immediate interaction targets. The second input is a **video visualizing the user's movement within the 3D scene**, which provides a global spatial context regarding the user's location within the scene and which furniture or region the user is moving toward. While the ego-centric video conveys local information about "what the user is looking at and interacting with," the 3D scene video provides complementary information about "where and within what spatial context the action takes place."

Qwen3-VL-14B receives both videos as input simultaneously, and the prompt is designed to classify each sequence into one of the 16 intents defined in Section C.1. The prompt includes (i) the definitions of the 16 intent categories, (ii) an explanation of the role of each of the two videos, and (iii) an instruction to select *unknown* when the sequence does not match any category. This dual-view approach effectively reduces the ambiguity that arises when using only a single ego-centric viewpoint (e.g., the confusion between merely standing in front of an object and actually intending to interact with it), thereby enabling the assignment of more consistent intent labels at the sequence level.

D Architectural Details

This appendix complements Section 3 of the main paper by specifying token construction, descriptor composition, and the components of the motion prediction model D_b.

D.1 Scene Tokenization

This appendix details the construction of the candidate grounding token set $\mathcal{C} = \mathcal{C}^{\text{obj}} \cup \mathcal{C}^{\text{nav}}$ used by the prior (Section 3.1).

Object-instance tokens. Object instances are obtained by zero-shot 3D instance segmentation on the scene point cloud S via OpenIns3D [26], producing per-instance point-cloud masks $\boldsymbol{m}_i \subseteq S$ with semantic labels.

Forward-facing viewing cone. We restrict object candidates to those within the person's plausible field of attention during the observation window. We define a per-frame viewing cone $C(x_\tau)$ originating at the head with forward-facing direction determined by body orientation $\boldsymbol{R}_\tau$, half-angle α_{cone}, and radius r_{cone}. The aggregated cone over the observation window is

$$C(M_{1:t}) = \bigcup_{\tau=1}^{t} C(x_\tau). \tag{18}$$

An object instance with mask $\boldsymbol{m}_i$ is included in $\mathcal{C}^{\text{obj}}$ if $\boldsymbol{m}_i \cap C(M_{1:t}) \neq \emptyset$, i.e., at least one of its points falls within the aggregated cone. Hyperparameter values ($\alpha_{\text{cone}}, r_{\text{cone}}$) are in Appendix F.

Navigation-region token. The navigation token $\mathcal{C}^{\text{nav}}$ represents object-agnostic free space (e.g., open floor, doorways, corridors). Unlike object tokens, it is kept scene-global (not cone-filtered) so that the prior can still represent free-space motion when the observed motion is not yet aligned with any specific object. Its mask is the complement of all object masks within an interaction-relevant region.

Rationale for cone-based filtering. Filtering object candidates by the viewing cone restricts the prior π to objects the person could plausibly attend to, and removes distractors that are unreachable or out of sight. This is grounded in the observation that upcoming interactions are preceded by progressive head and body alignment toward the target [20, 33, 69]. The navigation token, kept scene-global, preserves the option of free-space navigation when no object interaction is in progress.

D.2 Geometric Descriptor ϕ_i

The geometric descriptor concatenates two groups of features:

$$\phi_i = \left[\, \phi_{\text{geo}}(g_i),\ \phi_{\text{rel}}(c_i, M_{1:t}) \,\right] \in \mathbb{R}^F, \quad F = 16.$$

$\phi_{\text{geo}}(c_i)$ encodes token-intrinsic spatial properties (centroid, oriented bounding-box size, height above the floor, vertical orientation of the contact normal). $\phi_{\text{rel}}(c_i, M_{1:t})$ encodes motion-relative quantities computed at the last observed frame: signed distance from the root to c_i, root-to-token approach angle, root-horizontal alignment, face-normal alignment, and torso alignment.

D.3 Encoders and Student Scoring Head

Motion encoder E_M. A 4-layer Transformer with hidden dimension 256 and 4 heads ingests the per-frame SMPL-X state of $M_{1:t}$ and outputs a sequence-level motion feature h_M.

Token encoder $E_{\mathcal{C}}$. For each token c_i, we extract a point-cloud feature from P_i with a shared **PointNet++** backbone, concatenate it with the semantic-label embedding and a small projection of $\phi_{\text{geo}}(g_i)$, and pass the result through a two-layer MLP (GELU) to produce $e_i \in \mathbb{R}^{d_e}$ with $d_e = 16$.

Student scoring head F_S^{grd}. A permutation-equivariant set encoder over candidate tokens. Given h_M and $\{\phi_i\}_{i=1}^K$:

1. Per-token MLP: $u_i = \text{MLP}_{\text{cand}}([\phi_i; h_M]) \in \mathbb{R}^{d_h}$, $d_h = 256$, GELU.
2. Masked mean and max pooling over candidates: $\bar{u}^{\text{mean}}, \bar{u}^{\text{max}} \in \mathbb{R}^{d_h}$.
3. Context projection: $\bar{c} = \text{MLP}_{\text{ctx}}([\bar{u}^{\text{mean}}; \bar{u}^{\text{max}}]) \in \mathbb{R}^{d_h}$.
4. Per-token logit: $F_S^{\text{grd}}(h_M, \phi_i) = \text{MLP}_{\text{head}}([u_i; \bar{c}]) \in \mathbb{R}$.

The logits are normalized with a masked softmax (Eq. 8 in the main paper) to form π^S.

D.4 Geometric Compatibility Score

This appendix details each alignment term and the weight values used in the geometric compatibility score s_i^{geo} defined in Eq. (6) of the main paper.

Each alignment term measures the cosine similarity between a motion-derived direction at the last observed frame and the direction from the root toward token c_i, normalized to $[0, 1]$:

- $\text{align}_{\text{face}}(\phi_i, M_{1:t})$: alignment between the head-facing direction and the direction toward c_i.
- $\text{align}_{\text{body}}(\phi_i, M_{1:t})$: alignment between the torso-facing direction and the direction toward c_i.
- $\text{align}_{\text{line}}(\phi_i, M_{1:t})$: alignment between the recent trajectory direction and the direction toward c_i.

We fix $w_{\text{face}} = 0.5$, $w_{\text{body}} = 2$, $w_{\text{line}} = 0.5$ across all experiments, set on a held-out validation split.

D.5 Motion Predictor D_b

The motion prediction model D_b takes the encoded past motion, the selected token i_b^*, and the 3D scene as inputs, and produces the future trajectory and full-body pose. We adopt a transformer-based design with two parallel branches—a *global motion predictor* that focuses on translation and approach direction, and a *local motion predictor* that focuses on body articulation—which share scene features but consume the selected token in different ways. The student and the privileged teacher use the same architecture; only the source of the grounding signal differs.

Scene encoding. The scene point cloud S is encoded with **PointNet++** to produce per-point features $\hat{S} \in \mathbb{R}^{|S| \times d_p}$ and a global scene descriptor $\hat{S}_{\text{global}} \in \mathbb{R}^{d_p}$. Object-centric tokens $\{c_i\}$ are obtained by running **OpenIns3D** zero-shot 3D instance segmentation on S ; the per-token point sets $\{P_i\}$ allow either branch to attend to the points belonging to a selected token.

Selected token as conditioning. Given the selected token i_b^*, the conditioning signal is the pair $(m_{i_b^*}, e_{i_b^*})$. The mask $m_{i_b^*}$ induces a *local saliency point cloud*—the points of S belonging to i_b^*—which isolates the geometry around the inferred grounding region. The embedding $e_{i_b^*}$ encodes the token's semantic and geometric identity and serves as a token-conditioning bias broadcast over time.

Global motion predictor. This branch produces the future global trajectory $\hat{r}_{t+1:T}^b$ (translation and orientation). It consumes the encoded motion h_M together with the full scene features $\hat{S}$ and the token embedding $e_{i_b^*}$. A cross-attention block lets the trajectory query attend to $\hat{S}$, while $e_{i_b^*}$ is added to the query as a target-conditioning bias, pulling translation and approach direction toward the inferred grounding region. Operating on the full scene rather than the masked subset lets this branch reason about free-space and obstacle layout along the path to the target.

Local motion predictor. This branch produces the future body pose $\hat{\theta}^b_{t+1:T}$. It consumes h_M together with the local saliency point cloud (i.e., $\hat{S}$ restricted by $m_{i^*_b}$). A cross-attention block lets the pose query attend to these masked scene features, so that local body articulation—contact configuration, body orientation at arrival, and joint posture—is shaped by the geometry of the grounding region itself rather than by the full scene. We additionally feed the global trajectory features back into the local branch through a lightweight fusion layer, so that translation and orientation predicted by the global branch are consistent with the body posture predicted here.

Stack and fusion. We use $L_g = 4$ stacked blocks in the global branch and $L_l = 4$ stacked blocks in the local branch, hidden dimension $d_h = 256$, 4 attention heads, dropout 0.1. A final motion decoder assembles the predicted poses and the planned trajectory into the SMPL-X future motion $\hat{Y}^b_{t+1:T} = \{\hat{r}^b_{t+1:T}, \hat{\theta}^b_{t+1:T}\}$.

Comparison to gaze-conditioned decoders. Prior gaze-conditioned scene-aware HMP models [38, 63] inject a per-frame 3D gaze point as the scene-conditioning signal, which most directly affects the trajectory branch. Our predictor instead consumes the selected token i^*_b, derived end-to-end from $(M_{1:t}, S)$ via the prior $\hat{\pi}$ and requiring no auxiliary inference-time signal. The selected token participates in *both* the global and the local branch, so the inferred grounding shapes trajectory and pose jointly, rather than only the trajectory endpoint.

D.6 Attention Block Specifications

Trajectory Predictor(TP) Each Trajectory Predictor applies self-attention to motion tokens, cross-attention to scene features, and an MLP fusion with object embeddings:

$$H' = \mathrm{SelfAttn}(H_M) \tag{19}$$

$$C^{\mathrm{TP}} = \mathrm{softmax}\big(Q' K_S^\top / \sqrt{d}\big) V_S \tag{20}$$

$$H^{\mathrm{TP}} = \mathrm{MLP}\big([H', C^{\mathrm{TP}}, h^{\mathrm{obj}}]\big) \tag{21}$$

where $Q' = W_Q H'$, $K_S = W_K F_S$, $V_S = W_V F_S$, $F_S \in \mathbb{R}^{N\times d}$ are PointNet++ scene features, and h^{obj} is the object embedding. We stack 3 such blocks.

Pose Predictor(PP) The pose branch first injects trajectory information into motion tokens, then performs scene cross-attention with the spatial importance map P^b added as an additive bias:

$$\tilde{H} = H_M + W_{\mathrm{ori}} \hat{R} + W_{\mathrm{trans}} \hat{t} \tag{22}$$

$$H'' = \mathrm{SelfAttn}(\tilde{H}) \tag{23}$$

$$C^{\mathrm{PP}} = \mathrm{softmax}\big(Q'' K_S^\top / \sqrt{d} + P^b\big) V_S \tag{24}$$

$$H^{\mathrm{PP}} = \mathrm{MLP}\big([H'', C^{\mathrm{PP}}]\big) \tag{25}$$

where $P^b \in \mathbb{R}^{T\times N}$ is the trajectory-aware spatial importance map (see Eq. 26).

Spatial importance decomposition.

$$P^b = f_{\mathrm{dist}}(S, \hat{r}^b) + f_{\mathrm{dir}}(S, \hat{r}^b) + f_{\mathrm{mlp}}(S, \hat{r}^b) + m_{\mathrm{obj}} \cdot f_{\mathrm{obj}}(S, O, \hat{r}^b) \tag{26}$$

where f_{dist} is a distance-based prior, f_{dir} captures forward-facing direction alignment, f_{mlp} is a learned spatial bias, and f_{obj} is an object-direction prior weighted by the object validity mask m_{obj}.

D.7 Teacher Pose Refiner

Architecture. The pose refiner Refine (Section 3.3) adjusts the initial pose $\tilde{\boldsymbol{\theta}}^T_{t+1:T}$ from D^T_{pose} using the textual intent $\mathcal{I}$ and the selected token's embedding $\boldsymbol{e}^T_{i^*_T}$:

$$\hat{\boldsymbol{\theta}}^T_{t+1:T} = \mathrm{Refine}\big(\tilde{\boldsymbol{\theta}}^T_{t+1:T}, \mathcal{I}, \boldsymbol{e}^T_{i^*_T}\big). \tag{27}$$

To preserve the geometrically grounded trajectory, Refine keeps the predicted translation anchored and allows only a small zero-initialized orientation residual $\Delta\boldsymbol{R}^T_{t+1:T}$, permitting limited body reorientation (e.g., turning slightly toward an object before sitting) without disrupting the global motion path. Since the student has no access to $\mathcal{I}$ at inference, Refine is teacher-only; the student's pose comes directly from D^S_{pose}.

Intent-alignment loss. Refine is supervised by an intent-alignment loss that pulls the refined pose toward $\mathcal{I}$ in a shared text–pose feature space:

$$\mathcal{L}_{\text{intent-pose}} = \lambda_{\text{end}}\, d\big(\psi_\theta(\hat{\boldsymbol{\theta}}_T^T),\, \psi_c(\mathcal{I})\big) + \lambda_{\text{seq}}\, d\big(\psi_\theta(\hat{\boldsymbol{\theta}}_{t+1:T}^T),\, \psi_c(\mathcal{I})\big), \tag{28}$$

where ψ_c is a text encoder, ψ_θ a pose-sequence encoder mapping body pose into the same latent space, and $d(\cdot,\cdot)$ a cosine-based distance. The first term enforces alignment at the destination frame—the interaction state described by $\mathcal{I}$—while the second propagates intent semantics across the sequence. The orientation residual $\Delta\boldsymbol{R}_{t+1:T}^T$ from Eq. 27 is additionally regularized via a residual penalty $\lambda_{\text{ori}}\,\|\Delta\boldsymbol{R}_{t+1:T}^T\|$ that keeps orientation close to the trajectory anchor. Hyperparameter values for λ_{end}, λ_{seq}, λ_{ori}, the encoders ψ_c and ψ_θ, and the cosine distance form are in Appendix F.

D.8 Loss Components and Optimization

Notation. We follow Section 3.1: the root motion at frame τ is $r_\tau = (t_\tau, R_\tau)$ with translation $t_\tau \in \mathbb{R}^3$ and orientation $R_\tau \in SO(3)$, and the body pose is θ_τ in SMPL-X parameterization [44]. Predicted counterparts are denoted with hats, $\hat{r}_\tau = (\hat{t}_\tau, \hat{R}_\tau)$ and $\hat{\theta}_\tau$. The pose refiner's orientation residual $\Delta\boldsymbol{R}_{t+1:T}^T$ (Appendix D.7) is composed with the predicted orientation as $\hat{\boldsymbol{R}}_\tau^T \leftarrow \hat{\boldsymbol{R}}_\tau^T \cdot \Delta\boldsymbol{R}_\tau^T$ in $SO(3)$.

Motion reconstruction loss. The motion reconstruction loss for branch $b \in \{T, S\}$ is

$$\mathcal{L}_{\text{mot}}^b = \lambda_{\text{joint}}\,\mathcal{L}_{\text{joint}} + \lambda_{\text{vp}}\,\mathcal{L}_{\text{vposer}} + \lambda_{\text{mpjpe}}\,\mathcal{L}_{\text{mpjpe}} + \lambda_{\text{vel}}\,\mathcal{L}_{\text{vel}} + \lambda_{\text{foot}}\,\mathcal{L}_{\text{foot}} + \lambda_{\text{sem}}\,\mathcal{L}_{\text{sem}}. \tag{29}$$

The components are: joint-level reconstruction $\mathcal{L}_{\text{joint}}$ on 3D joint positions; VPoser-latent regularization $\mathcal{L}_{\text{vposer}}$ that pulls predicted poses toward the VPoser prior [44]; non-root MPJPE $\mathcal{L}_{\text{mpjpe}}$; inter-frame velocity consistency $\mathcal{L}_{\text{vel}}$; foot-contact penalty $\mathcal{L}_{\text{foot}}$ that suppresses sliding during ground contact; and semantic-prompt alignment $\mathcal{L}_{\text{sem}}$. Definitions of individual loss terms follow [38, 63].

Optimized parameter sets.

- Trajectory pass updates $\{E_M^T, E_{\mathcal{G}}^T, D^T\}$ with $\mathcal{L}_{\text{mot}}^T$.
- Pose-refinement pass freezes $\{E_M^T, E_{\mathcal{G}}^T, D_{\text{traj}}^T\}$ and updates R^T (with $\Delta\boldsymbol{R}^T$) using $\mathcal{L}_{\text{intent-pose}}$ + pose-reconstruction + residual penalty.
- Student training updates $\{E_M^S, E_{\mathcal{G}}^S, F_S^{\text{grd}}, D^S\}$ using $\mathcal{L}^S$ (Eq. 15).

E Auxiliary Distillation Losses

The prior distillation $\mathcal{L}_{\text{KD}}^z$ in Eq. 13 of the main paper is a forward KL between the privileged distribution π^T and the student π^S. To sharpen the rubric-induced top-1 and preserve the rubric ordering on salient candidates, we additionally consider three optional auxiliary terms; these contribute small additional improvements and are reported here for completeness.

Rubric-argmax cross-entropy. With $i^*_{\text{rubric}} = \arg\max_i \pi_i^T$ (the rubric argmax *before* the geometry score is added),

$$\mathcal{L}^{\text{rCE}} = -\log \pi^S_{i^*_{\text{rubric}}}. \tag{30}$$

The target is i^*, the rubric argmax computed *before the geometry score is added* (Appendix D.4), rather than the geometry-adjusted argmax i_T^*. Hence the geometry-score refinement does not enter as supervision.

Top-k pairwise ranking margin loss. Let $\mathcal{P}$ be the set of pairs (i, j) from the rubric top-k with $\pi_i^T - \pi_j^T > \tau_q$. With margin η,

$$\mathcal{L}^{\text{rank}} = \frac{1}{|\mathcal{P}| \vee 1} \sum_{(i,j)\in\mathcal{P}} \big(\eta - (z_i^S - z_j^S)\big)_+, \tag{31}$$

where z_i^S is the student logit before softmax.

Top-k pair-gap Huber loss. On the same pair set $\mathcal{P}$, with a numerical floor $\hat{\pi} \leftarrow \max(\hat{\pi}, 10^{-8})$ before $\log$, we Huber-regress the student log-odds gap toward the teacher gap with threshold $\beta = 1$:

$$\mathcal{L}^{\mathrm{pg}} = \frac{1}{|\mathcal{P}| \vee 1} \sum_{(i,j)\in\mathcal{P}} h_\beta\big((z_i^S - z_j^S) - (\log \pi_i^T - \log \pi_j^T)\big). \tag{32}$$

Use in main results. Our main observation-only objective is $\mathcal{L}_S = \mathcal{L}^S_{\mathrm{mot}} + \lambda_z \mathcal{L}^z_{\mathrm{KD}} + \lambda_Y \mathcal{L}^Y_{\mathrm{KD}}$, i.e., $\lambda_{\mathrm{rCE}} = \lambda_{\mathrm{rank}} = \lambda_{\mathrm{pg}} = 0$. The auxiliary terms can be enabled at small weights ($\lambda_{\mathrm{rCE}} = 0.1$, $\lambda_{\mathrm{rank}} = 0.2$, $\lambda_{\mathrm{pg}} = 0.5$) when sharper top-1 grounding is preferred over mass-covering soft distributions.

F Hyperparameters, Training Schedule, and Compute

F.1 Tensor Shapes

For batch size B, maximum candidate count K, and $F = 16$ geometric features:

Quantity	Shape	Notes
Geometric input Φ	(B, K, F)	z-score normalized using train statistics
Validity mask	(B, K)	boolean
Token embedding E	(B, K, d_e)	$d_e = 16$
Point-cloud mask m	$(B, K, \lvert S\rvert)$	binary
Rubric total s^{rubric}	(B, K)	integer, $[2, 10]$
Teacher prior π^T	(B, K)	on the simplex Δ^{K-1}
Student prior π^S	(B, K)	on the simplex Δ^{K-1}
Predicted future $\hat{Y}$	$(B, T-t, d_Y)$	SMPL-X trajectory + pose

Table 4: Tensor shapes throughout the SGIP pipeline.

F.2 Hyperparameters

Encoders and motion prediction model. E_M: 4-layer Transformer, hidden dimension 256, 4 attention heads. $E_{\mathcal{G}}$: PointNet++ backbone with 2 set-abstraction layers, output dimension $d_e = 16$. D_b: $L_t = 4$ stacked blocks in the trajectory predictor and $L_p = 4$ stacked blocks in the pose predictor, hidden dimension $d_h = 256$, 4 attention heads, dropout 0.1.

Student scoring head. $d_h = 256$; per-token MLP and head are 2-layer MLPs with GELU.

Loss weights. Trajectory $\lambda_{\mathrm{traj}} = 1.0$, pose $\lambda_{\mathrm{pose}} = 1.0$, velocity $\lambda_{\mathrm{vel}} = 0.1$. Distillation $\lambda_z = 1.0$, $\lambda_Y = 0.5$ (with $\lambda^{\mathrm{KD}}_{\mathrm{traj}} = 1.0$, $\lambda^{\mathrm{KD}}_{\mathrm{pose}} = 0.5$). Auxiliary losses (Appendix E) are disabled in main results.

F.3 Training Schedule

Stage 1 (privileged teacher). AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 8, 100 epochs. Cosine learning-rate schedule with 5-epoch linear warm-up.

Stage 2 (observation-only student). The privileged branch is frozen. AdamW, learning rate 10^{-3}, weight decay 10^{-4}, batch size 32, 200 epochs, same schedule. Both π^T and $\hat{Y}^T_{t+1:T}$ are stop-gradient targets.

F.4 Compute

All experiments run on a single NVIDIA RTX 3090 (24 GB). Each training stage takes approximately 2 hours; the LLM scoring step is run once over the full GIMO training set in approximately 4 hours and cached as CSV. No LLM call occurs during training updates or inference. Random seed is 7.

G Additional Ablation Studies

This appendix complements Section 4.3 of the main paper with extended setup and analysis for two of the ablations referenced there: the accuracy of the predicted scene-grounded interaction prior on the GIMO-annotated interaction target (Table 2c), and the comparison between hard top-1 and soft top-k grounding-token aggregation (Table 2b). The finer-grained decomposition of the distillation losses is discussed in Section 4.3 of the main paper.

G.1 Grounding Token Selection Accuracy

Since $\pi^{\mathcal{T}}$ serves as the distillation target for $\pi^{\mathcal{S}}$ (Eq. 13), a natural diagnostic is whether the two priors concentrate mass on the GIMO-annotated interaction token. This annotation is used here only for evaluation: it never enters training (Table 3) and is never accessed at inference.

Candidate set statistics. We use OpenIns3D [26] to extract object instances per GIMO scene, augmented by a single navigation-region token (Appendix D.1). We match GIMO ground-truth targets to OpenIns3D proposals by IoU on point membership (threshold 0.5); cases where zero-shot segmentation fails to recover the target bound the achievable top-1 accuracy and are acknowledged in the limitations (Section 5).

Top-k accuracy. Table 2c reports top-k accuracy on the matched ground-truth target for four selection strategies: a rubric-only baseline using the LLM score alone, a geometric-only baseline using the motion-relative descriptor $\boldsymbol{\phi}_{\text{rel}}$ alone, the deployed student $\pi^{\mathcal{S}}$, and the privileged teacher $\pi^{\mathcal{T}}$. The unimodal baselines plateau at $54\%/51\%$ top-1, reflecting two complementary failure modes: the LLM score cannot disambiguate same-category instances (e.g., several chairs), while the geometric score is intent-blind and confuses intent-equivalent objects (e.g., a cup vs. a bottle for *drink*). The teacher's 86% top-1 quantifies the headroom from fusing both signals, and the student retains 76% top-1 from motion and scene alone, directly evidencing that distillation transfers intent-aware grounding to an observation-only model.

G.2 Hard vs. Soft Grounding Token Aggregation

The motion predictor consumes a single grounding token i_b^* via its mask $\boldsymbol{m}_{i_b^*}$ and embedding $\boldsymbol{e}_{i_b^*}$ (Eqs. 10, 12). A natural alternative is to aggregate the top-k tokens softly, replacing the hard argmax (Eq. 9) with a probability-weighted mixture:

$$\bar{\boldsymbol{m}} = \sum_{i \in \mathcal{T}_k} \tilde{\pi}_i^{\mathcal{S}}\, \boldsymbol{m}_i, \qquad \bar{\boldsymbol{e}} = \sum_{i \in \mathcal{T}_k} \tilde{\pi}_i^{\mathcal{S}}\, \boldsymbol{e}_i, \tag{33}$$

where $\mathcal{T}_k = \mathrm{TopK}_k(\pi^{\mathcal{S}})$ and $\tilde{\pi}_i^{\mathcal{S}} = \pi_i^{\mathcal{S}} / \sum_{j \in \mathcal{T}_k} \pi_j^{\mathcal{S}}$. The predictor $D^{\mathcal{S}}$ then consumes $(\bar{\boldsymbol{m}}, \bar{\boldsymbol{e}})$ in place of $(\boldsymbol{m}_{i_{\mathcal{S}}^*}, \boldsymbol{e}_{i_{\mathcal{S}}^*})$.

Soft aggregation provides a smoother conditioning signal under candidate ambiguity, but it also dilutes the geometry of the selected region by blending in less-likely tokens. As reported in Table 2b, hard top-1 retains a clear advantage across all metrics, validating the discrete grounding-selector design for single-future prediction. Soft aggregation may be more appropriate when the goal shifts to diverse-future generation, where preserving the multimodal structure of $\pi^{\mathcal{S}}$ is desirable; we leave this to future work (Section 5).

H Additional Qualitative Results

This appendix provides two additional qualitative analyses that complement Figure 4 of the main paper: (i) the effect of privileged distillation on the student's predicted motion (Figure 5), and (ii) the multimodal futures induced by conditioning on the top-k grounding tokens of $\pi^{\mathcal{S}}$ (Figure 6).

H.1 Effect of Distillation on Student Predictions

The qualitative comparison aligns with the quantitative gap in Figure 5 and underscores the advantage of SGIP over the SIF3D baselines. SIF3D without gaze drifts toward an incorrect region with

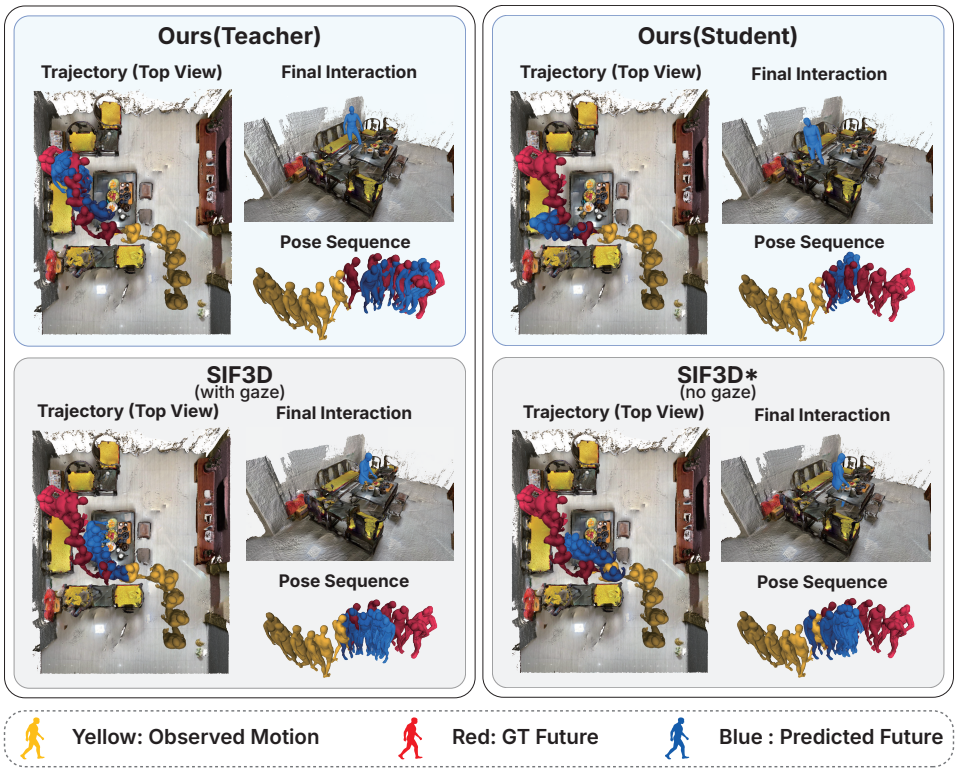


Figure 5: **Qualitative comparison across baselines and SGIP variants.** For one representative cluttered scene from the GIMO test set, we compare four methods given the same observed motion prefix: **(a)** SIF3D [38] with gaze, **(b)** SIF3D without gaze, **(c)** the privileged SGIP teacher ($\text{SGIP}^{\dagger}_{\text{priv}}$), and **(d)** the deployed observation-only SGIP student. Each panel shows the top-down trajectory (top) together with three keyframe poses along the predicted future (bottom). Without gaze, SIF3D drifts toward an incorrect region and produces an inconsistent arrival pose; the gaze-informed SIF3D recovers the correct direction but still over- or under-shoots the target. Both SGIP variants ground the trajectory on the correct sofa and produce a coherent sit-down pose at arrival, with the observation-only student closely matching the privileged teacher despite using neither gaze nor textual intent at inference.

an inconsistent arrival pose, while gaze-augmented SIF3D recovers the rough direction but still over- or under-shoots the target—indicating that gaze alone is insufficient to disambiguate the interaction target in cluttered scenes. In contrast, both SGIP variants ground the trajectory on the correct chair and produce a coherent sit-down pose at arrival. The observation-only student closely tracks the privileged teacher despite using neither gaze nor textual intent at inference, confirming that intent-aware grounding is successfully transferred through privileged distillation. This advantage is most pronounced in scenes with multiple same-category instances (e.g., two chairs): the SIF3D baselines frequently approach a non-target instance, whereas SGIP reliably arrives at the ground-truth target with a coherent body orientation, yielding both more accurate trajectory grounding and more plausible interaction poses than all baselines.

H.2 Multimodal Futures from Top-k Grounding Tokens

A central design claim of SGIP is that the predicted prior π^S is a structured distribution over candidate grounding tokens, rather than a one-hot indicator of a single interaction target. To probe this claim, we condition the motion predictor D^S on each of the top-3 grounding tokens of π^S in turn and visualize the resulting trajectory and arrival pose. The student backbone is held fixed; only the selected token (m_i, e_i) fed into D^S varies across the three columns.

Two observations follow from Figure 6. First, the top-2 and top-3 candidates correspond to semantically meaningful alternative interactions: in scenes with multiple chair instances, top-2 typically

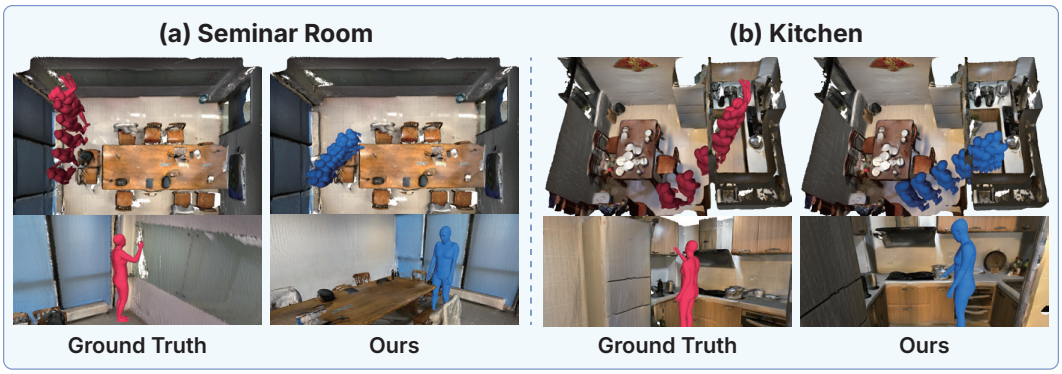


Figure 6: **Multimodal futures from top-k grounding tokens.** For two representative cluttered scenes from the GIMO test set, we condition the motion predictor D^S on each of the top-3 grounding tokens of the student prior π^S in turn and visualize the resulting trajectory (top row of each block) and final-frame arrival pose (bottom row); the rightmost column shows the ground-truth future motion. The student backbone is held fixed; only the selected token (m_i, e_i) fed into D^S varies across the three columns. **(a)** A scene with two chair instances and a sofa: the predicted prior is $\pi^S = \{\text{chair-A} : 0.42, \text{chair-B} : 0.31, \text{sofa} : 0.18, \dots\}$. Top-2 selects the alternative chair-B with a longer but coherent approach, and top-3 selects the sofa with a correspondingly different arrival pose; the top-1 prediction matches the ground truth. **(b)** A scene with mixed object types: $\pi^S = \{\text{sofa} : 0.39, \text{chair} : 0.33, \text{stool} : 0.17, \dots\}$. Each top-$k$ candidate corresponds to a semantically distinct, scene-consistent interaction target rather than a degenerate variant of the top-1; in this case the second-ranked chair coincides with the ground truth, while top-1 (sofa) and top-3 (stool) remain individually plausible. These results indicate that π^S preserves genuine multimodal structure and that D^S has learned a token-conditional motion distribution rather than a token-agnostic average, supporting the diverse-future extension discussed in Section 5.



Figure 7: **Zero-shot 3D instance segmentation by OpenIns3D [26].** For two representative GIMO scenes, we visualize the per-instance point-cloud masks $\boldsymbol{m}_i$ produced by OpenIns3D on the reconstructed scene point cloud S. Each color denotes a distinct object instance $c_i \in \mathcal{C}^{\text{obj}}$. Same-category objects (e.g., multiple chairs) are correctly separated as distinct instances, a precondition for our scene-grounded interaction prior to disambiguate among them via the geometric compatibility score (Appendix D.4).

selects the second chair with a coherent (if longer) approach; in scenes with mixed object types, top-2 and top-3 may select a different category entirely (e.g., a sofa instead of a chair) with a correspondingly different arrival pose. Second, the motion predictor produces trajectory and pose sequences that are individually plausible for each conditioned token, even when that token is not the ground truth. This indicates that D^S has learned a token-conditional motion distribution rather than a token-agnostic average, and that the multimodality captured by π^S is reflected at the output level rather than only at the level of token logits.

These qualitative results support the future-work direction discussed in Section 5: instead of committing to a single grounding token at inference, retaining the full predicted prior π^S and sampling

multiple plausible futures from it is a natural extension that respects the inherent multimodality of scene-aware human motion prediction.

I Zero-Shot 3D Instance Segmentation Results

Figure 7 shows the zero-shot 3D instance segmentation output of OpenIns3D [26] on representative GIMO scenes. The resulting per-instance masks $\{\boldsymbol{m}_i\}$ and semantic labels form the object-centric candidate set $\mathcal{C}^{\text{obj}}$ used by the scene-grounded interaction prior (Appendix D.1).

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [YES]

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